Formal Methods in Software Engineering

Boolean Satisfiability, Spring 2026

Konstantin Chukharev

SAT Problem

Boolean Satisfiability Problem (SAT)

A propositional formula is *satisfiable* if it has a *model* — an assignment of truth values that makes it true.

Definition 1 (Boolean Satisfiability (SAT)): Given a propositional formula φ over variables $X = \{x_1, ..., x_n\}$, decide whether

$$\exists \nu : X \to \{0, 1\}. \quad \nu \vDash \varphi$$

SAT (decision): does a model exist? **Functional SAT**: also return a concrete ν.

SAT solvers work with formulas in CNF: a conjunction of *clauses*, each a disjunction of *literals*.

Example: Formula: $(x_1 \vee \neg x_2) \wedge (\neg x_1 \vee x_3) \wedge (x_2 \vee x_3)$.

- Assignment $\{x_1 = 1, x_2 = 0, x_3 = 1\}$: every clause is satisfied — this is a *model*.
- Assignment $\{x_1 = 0, x_2 = 1, x_3 = 0\}$: **not** a *model*.
  - clause $(x_2 \vee x_3) = (1 \vee 0) = 1$ ✓
  - clause $(\neg x_1 \vee x_3) = (1 \vee 0) = 1$ ✓
  - clause $(x_1 \vee \neg x_2) = (0 \vee 0) = 0$ ✗

The Cook–Levin Theorem

Theorem 1 (Cook–Levin (Cook 1971, Levin 1973)): SAT is NP-complete: it is in NP, and *every* problem in NP can be reduced to SAT in polynomial time.

Proof idea: Every NP problem has a polynomial-time verifier V (a Turing machine). We encode V's execution as a Boolean formula:

Variables (for T steps, S cells):

- $q_{t,s}$: machine in state s at step t
- $h_{t,p}$: head at position p at step t
- $c_{t,p,\sigma}$: cell p has symbol σ at step t

Clauses enforce valid computation:

- *Initial config* — input on tape
- *Transition function* — δ as implications
- *Acceptance* — accepting state reached

The resulting formula φ is satisfiable iff V accepts on some certificate. The reduction is polynomial: $O(T^2)$ clauses, where $T = p(n)$ is the verifier's runtime.

SAT Encoding Methodology

To reduce a search problem to SAT:

1. **Define variables** for each binary choice in the problem.
2. **Encode constraints** as propositional clauses.
3. **Convert to CNF** if needed (use Tseitin to avoid exponential blowup).
4. **Run a SAT solver** to obtain a model or an UNSAT proof.

Example: **Graph k-colorability:** can we color the vertices of $G = (V, E)$ with k colors so adjacent vertices receive different colors?

- **Variables:** $c_{v,i}$ = "vertex v gets color i", for $v \in V$, $i \in \{1, \ldots, k\}$.
- **EO per vertex:** each vertex gets exactly one color.
- **Edge constraint:** for each $(u, v) \in E$ and color i: $\left(\neg c_{u,i} \vee \neg c_{v,i}\right)$.

If the solver returns SAT, reading off the values of $c_{v,\cdot}$ gives the coloring.

Encoding Patterns: At-Least-One & At-Most-One

Encoding cardinality constraints is a recurring task.

Definition 2: *At least one* (ALO) of $x_1, ..., x_n$ is true:

$$(x_1 \vee x_2 \vee ... \vee x_n)$$

One clause with n literals.

Definition 3: *At most one* (AMO) of $x_1, ..., x_n$ is true. **Pairwise encoding:** for each pair $i < j$,

$$(\neg x_i \vee \neg x_j)$$

One clause per pair: $\binom{n}{2}$ clauses total.

Encoding Patterns: At-Least-One & At-Most-One [2]

Example: AMO on $\{x_1, x_2, x_3\}$: three clauses $(\neg x_1 \vee \neg x_2) \wedge (\neg x_1 \vee \neg x_3) \wedge (\neg x_2 \vee \neg x_3)$. The assignment $x_1 = 1, x_2 = 1, x_3 = 0$ falsifies the first clause: $(0 \vee 0) = 0$. Any assignment with at most one true variable satisfies all three.

Note: Pairwise AMO produces $O(n^2)$ clauses. For large n, *commander–variable* or *logarithmic* encodings achieve $O(n)$ clauses using auxiliary variables.

Encoding Patterns: Exactly-One & Implications

Definition 4: *Exactly one* (EO) of $x_1, \ldots, x_n$ is true: ALO $\wedge$ AMO combined.

$$\underbrace{(x_1 \vee \ldots \vee x_n)}_{\text{ALO}} \wedge \underbrace{\bigwedge_{i<j} (\neg x_i \vee \neg x_j)}_{\text{AMO}}$$

Common encoding primitives:

- **Implication:** $a \to b$ becomes $(\neg a \vee b)$ — one clause.
- **If-then-else:** $\text{ite}(c, t, e)$ becomes $(\neg c \vee t) \wedge (c \vee e)$ — two clauses.
- **Mutual exclusion:** "at most one of $x_1, \ldots, x_n$" — use AMO.
- **Channeling:** link two groups of variables, *e.g.*, $x_{i,j} \iff y_{j,i}$.

Common mistake: Encoding ALO without AMO. Without the pairwise clauses, the solver is free to set *multiple* variables true — the encoding is too weak to rule out invalid solutions.

Encoding Patterns: Summary

Pattern	Clauses	Aux Vars	When to Use
ALO$(x_1, \ldots, x_n)$	1	0	Something must be chosen
AMO pairwise	$\binom{n}{2}$	0	At most one choice ($n \leq 10$)
AMO commander	$O(n)$	$O(n)$	At most one choice ($n > 10$)
EO = ALO + AMO	$1 + \binom{n}{2}$	0	Exactly one choice
Implication $a \to b$	1	0	Dependency between choices
If-then-else	2	0	Conditional assignment

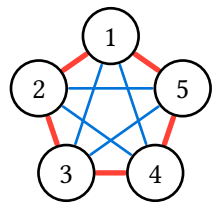
SAT Encodings

Example: Graph Coloring

Graph $G = (V, E)$: vertices V, edges E (unordered pairs). K_n — the complete graph on n vertices (every pair connected).

Problem: Color the *edges* of K_n using k colors with *no monochromatic triangle*. What is the largest n for which this is possible?

- For $k = 1$: $n = 2$ (only 1 edge).
- For $k = 2$: $n = 5$ (see diagram on the right).
- For $k = 3$: $n = 16$ (requires a SAT solver to verify).



Modelling Graph Coloring as SAT

1. **Variables:** For each edge e and color $c \in \{1, \ldots, k\}$, define e_c ("edge e has color c").
2. **Constraints:**

   Each edge gets *exactly one* color (ALO + AMO):

   $$(e_1 \vee e_2 \vee e_3) \wedge \neg(e_1 \wedge e_2) \wedge \neg(e_1 \wedge e_3) \wedge \neg(e_2 \wedge e_3)$$

   No monochromatic triangle — for each triangle (e, f, g) and color c:

   $$\neg(e_c \wedge f_c \wedge g_c)$$

3. **CNF:** The constraints above are already (close to) CNF.
4. **Solve:** Increase n until the formula becomes UNSAT.

The EO constraint on edge colors is exactly the ALO + AMO pattern from the previous section — applied to the set of color variables for each edge.

DIMACS CNF Format

SAT solvers use the *DIMACS CNF* format — a standard text representation:

```
c This is a comment
p cnf 4 3
1 2 -3 0
-1 3 0
2 3 4 0
```

- `p cnf <vars> <clauses>` — header
- Variables: positive integers $1, 2, \ldots$
- Negation: prefix with `-`
- Each clause ends with `0`
- Comments start with `c`

Run a solver:

```
cadical formula.cnf
```

If SAT, the solver outputs a *model* (a line of `v` values). If UNSAT, a DRAT *proof certificate* can be requested with `--proof`.

Code: Graph Coloring SAT Encoding

```
n = 17
k = 3
m = n * (n - 1) // 2

edges = {}
for u in range(1, n + 1):
    for v in range(u + 1, n + 1):
        edges[(u, v)] = len(edges) + 1

def color(e, c):
    return (e - 1) * k + c

clauses = []
for e in range(1, m + 1):
    # ALO: at least one color per edge
    clauses.append([
      color(e, c) for c in range(1, k + 1)
    ])
    # AMO: at most one color per edge
    for c1 in range(1, k + 1):
        for c2 in range(c1 + 1, k + 1):
            clauses.append([
              -color(e, c1), -color(e, c2)
            ])
# No monochromatic triangles
for v1 in range(1, n + 1):
    for v2 in range(v1 + 1, n + 1):
        for v3 in range(v2 + 1, n + 1):
            e12 = edges[(v1, v2)]
            e23 = edges[(v2, v3)]
            e13 = edges[(v1, v3)]
            for c in range(1, k + 1):
                clauses.append([
                  -color(e12, c),
                  -color(e23, c),
                  -color(e13, c)
                ])
# Output DIMACS CNF
print(f"p cnf {color(m, k)} {len(clauses)}")
for clause in clauses:
    print(" ".join(map(str, clause)) + " 0")
```

Example: N-Queens

Place n queens on an $n \times n$ board so no two attack each other.

Variables: $q_{i,j}$ = "queen on row i, column j" (n^2 variables).

Constraints:

- **EO per row:** exactly one queen in each row i: ALO$\left(q_{i,1}, \ldots, q_{i,n}\right)$ + AMO.
- **AMO per column:** at most one queen in each column j: AMO$\left(q_{1,j}, \ldots, q_{n,j}\right)$.
- **AMO per diagonal:** at most one queen on each diagonal and anti-diagonal.

Size: n^2 variables, $O(n^3)$ clauses (pairwise AMO on each line).

Note: N-Queens is a classic SAT benchmark. For $n = 1000$, the encoding has 10^6 variables and $\sim 10^9$ clauses — but modern solvers handle it in seconds.

Example: Pigeonhole Principle (Sketch)

Place $n+1$ pigeons into n holes, at most one pigeon per hole.

Variables: $p_{i,j}$ = "pigeon i goes into hole j" $\quad (n(n+1)$ variables).

Constraints:

- **ALO per pigeon:** each pigeon gets a hole: $\left(p_{i,1} \vee \ldots \vee p_{i,n}\right)$.
- **AMO per hole:** each hole has at most one pigeon: $\left(\neg p_{i,j} \vee \neg p_{k,j}\right)$ for $i \neq k$.

This formula is **always UNSAT** ($n+1$ pigeons cannot fit in n holes).

Proof complexity: Resolution proofs of PHP_{n+1}^{n} require exponentially many steps (Haken, 1985). DPLL implicitly constructs resolution proofs, so it struggles here. CDCL with learned clauses can do better — but pigeonhole remains hard for all known solvers.

Encodings: Key Takeaways

The SAT encoding recipe:

1. Identify the *choices* in your problem ⇒ propositional variables.
2. Express *validity conditions* using ALO, AMO, EO, implication patterns.
3. Convert to CNF (usually straightforward; use Tseitin if needed).
4. Feed to a SAT solver and interpret the result.

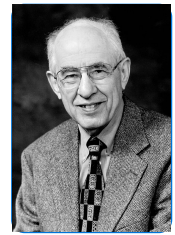
The expressiveness comes from NP-completeness: any polynomially verifiable property encodes into SAT.
The efficiency comes from the solvers themselves: modern CDCL handles *billions* of clauses.

Algorithms for SAT

Davis–Putnam Algorithm

The first algorithm for SAT was proposed by Martin Davis and Hilary Putnam in 1960 [1].

Martin Davis



Hilary Putnam

Satisfiability-preserving simplification rules:

1. **Unit propagation** — propagate forced assignments.
2. **Pure literal elimination** — remove variables appearing with one polarity.
3. **Resolution** (variable elimination) — resolve away a variable.

The original DP algorithm uses resolution, which can *increase* formula size. DPLL (1962) replaces resolution with *splitting* (backtracking search), which is far more practical.

Henceforth, formulas are represented in **CNF**: a set of clauses, each a set of literals.

Unit Propagation Rule

Definition 5 (Unit clause): A *unit clause* is a clause with a single literal.

Suppose (p) is a unit clause. Recall that $\overline{p}$ denotes the complement literal: $\overline{p} = \begin{cases} \neg p \text{ if } p \text{ is positive} \\ p \text{ if } p \text{ is negative} \end{cases}$

Unit propagation:

- Assign p to true.
- Remove all clauses containing p (they are satisfied).
- Remove $\overline{p}$ from all remaining clauses (it is falsified).

Example: Consider $(A \vee B) \wedge (A \vee \neg B) \wedge (\neg A \vee B) \wedge (\neg A \vee \neg B) \wedge (A)$.

- The unit clause (A) forces $A = 1$.
- Remove clauses with A; remove $\neg A$ from the rest:

  ~~$(A \vee B)$~~ $\wedge$ ~~$(A \vee \neg B)$~~ $\wedge$ $(\neg A \vee B)$ $\wedge$ ~~$(\neg A \vee \neg B)$~~ $\wedge$ ~~(A)~~

- Result: $(B) \wedge (\neg B)$ — still unsatisfiable.

Pure Literal Rule

Definition 6: A literal p is *pure* if it appears in the formula only positively or only negatively.

Pure literal elimination:

- Assign the pure literal to true.
- Remove all clauses containing it (they are now satisfied).

Example: $(A \vee B) \wedge (A \vee C) \wedge (B \vee C)$.

- Literal A is pure (appears only positively).
- Assign $A = 1$, remove clauses containing A: result is $(B \vee C)$.

Note: Unit propagation is a *forced* assignment (no choice). Pure literal elimination is a *safe* assignment (any model can be extended). Both reduce the formula without branching.

Davis–Putnam–Logemann–Loveland (DPLL)

Input: set of clauses S
Output: *satisfiable* or *unsatisfiable*

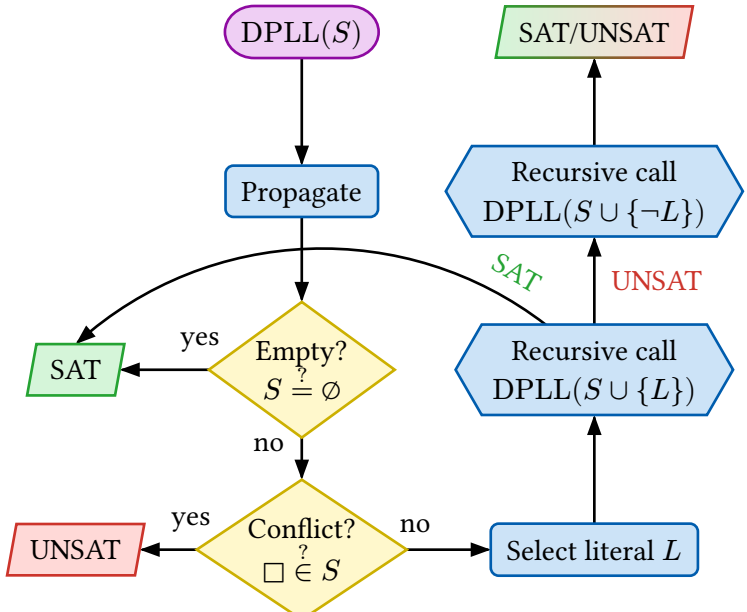
1 $S :=$ propagate(S)
2 **if** S is empty **then**
  return *satisfiable*
3 **if** S contains the empty clause $\square$ **then**
  return *unsatisfiable*
4 $L :=$ select_literal(S)
5 **if** DPLL$(S \cup \{L\}) =$ *satisfiable* **then**
  return *satisfiable*
6 **else**
  return DPLL$(S \cup \{\neg L\})$
7 **end**

DPLL [2] replaces resolution with *splitting*: pick a literal, assert it, propagate, recurse; on failure, assert its negation.

DPLL is *complete*: it always terminates and finds a satisfying assignment iff one exists.

The search forms a binary tree where each internal node is a literal choice and leaves are SAT or UNSAT.

DPLL Flowchart



Worked DPLL Trace

Consider: $(A \vee B) \wedge (\neg A \vee C) \wedge (\neg B \vee \neg C) \wedge (A \vee \neg C)$ — 4 clauses, 3 variables.

Step	Action	Assignment	Clauses	Status
1	Decide $A = 1$	$A = 1$	$(\neg A \vee C) \Rightarrow (C)$; others simplified	
2	Unit prop $C = 1$	$A = 1, C = 1$	$(\neg B \vee \neg C) \Rightarrow (\neg B)$	
3	Unit prop $B = 0$	$A = 1, C = 1, B = 0$	Check $(A \vee B)$: satisfied	SAT ✓

Model: $\nu = \{A = 1, B = 0, C = 1\}$. Verify: all clauses satisfied.

Note: In this trace, DPLL found a model without backtracking. On structured hard instances (pigeonhole, parity), the search tree can have exponentially many branches.

From DPLL to CDCL

DPLL has a fundamental limitation: **chronological backtracking**.

When a conflict occurs, DPLL backtracks to the most recent decision — even if that decision was irrelevant to the conflict. This forces the solver to re-explore search spaces that cannot possibly contribute to a solution.

DPLL:

- Backtrack to the previous decision
- Undo it, try the other value
- No "memory" of *why* the conflict occurred
- May repeat the same mistake in a different subtree

CDCL:

- Analyze the conflict: which prior decisions caused it?
- Learn a new clause that excludes this combination.
- Backjump to the earliest relevant decision level.
- The same conflict assignment is now ruled out by the learned clause.

CDCL answer: analyze the implication graph to identify which decisions caused the conflict, learn a clause encoding that fact, and jump directly to the responsible level.

DPLL: Key Takeaways

DPLL = backtracking + unit propagation + pure literal.

- Decides a variable, propagates consequences, recurses.
- Complete: always finds a solution or proves UNSAT.
- Worst case: $O(2^n)$ — explores the full binary decision tree.
- The backbone of *all* modern SAT solvers.

Note: DPLL can be formalized as a *transition system* with rules for unit propagation, decisions, and backtracking [3]. This abstract framework extends naturally to CDCL via Learn and Backjump rules.

Conflict-Driven Clause Learning

Implication Graph

During propagation, each forced assignment has a *reason* — the clause that caused it.
The implication graph makes these dependencies explicit.

Definition 7 (Implication Graph): A labeled directed acyclic graph where:

- **Nodes** represent assigned literals, labeled with their decision level (1, 2, 3, ...).
- **Edges** from a clause C point to the propagated literal ℓ with label C ("clause C forced ℓ").
- **Source nodes** correspond to decisions (no incoming edges, marked ■).
- **Sink node** κ is a special conflict node, with incoming edges from literals in the conflicting clause.

Example: Consider the formula from the next slide: $c_1 = (x_1 \vee x_2)$, $c_3 = (\neg x_2 \vee x_3)$, $c_4 = (\neg x_3 \vee x_4)$, $c_5 = (\neg x_3 \vee \neg x_4)$. At decision level 1, assign $x_1 := 0$.

Unit propagations: c_1 forces $x_2 := 1$; c_3 forces $x_3 := 1$; c_4 forces $x_4 := 1$; $c_5 = (0 \vee 0)$ — conflict.

All four literals are at level 1. Implication graph (edges labeled by forcing clause):

$$\overline{x}_1 \xrightarrow{c_1} x_2 \xrightarrow{c_3} x_3 \xrightarrow{c_4} x_4 \xrightarrow{c_5} \kappa$$

with a second edge $x_3 \xrightarrow{c_5} \kappa$ (both $\neg x_3$ and $\neg x_4$ are in the conflict clause).

Conflict Analysis

When a conflict occurs, CDCL traces the implication graph backward to derive a clause that explains the conflict.

Definition 8 (1-UIP (Unique Implication Point)): The *first unique implication point* (1-UIP) is the node at the current decision level d that is closest to κ and dominates *all* paths from the level-d decision to κ.

Learned clause: negate the 1-UIP literal, plus the negations of all prior-level literals that have edges into the conflict side of the cut.

Example: From the previous slide.

- Paths from $\overline{x}_1$ to κ: $\overline{x}_1 \to x_2 \to x_3 \to x_4 \to \kappa$ and $\overline{x}_1 \to x_2 \to x_3 \to \kappa$.
- Both pass through x_3; neither requires x_4 on both.
- So x_3 dominates all paths — it is the 1-UIP.

Apply the resolution procedure:

- Start with conflict clause $c_5 = (\neg x_3 \vee \neg x_4)$.
- Two level-1 literals.

Conflict Analysis [2]

- Resolve on x_4 with its reason $c_4 = (\neg x_3 \vee x_4)$: the resolvent is $(\neg x_3)$.
- One level-1 literal remains.
- **Learned clause: $(\neg x_3)$.**

No prior-level literals appear, so backjump to level 0 and propagate $x_3 := 0$.

The learned clause is added permanently to the clause database.

Each learned clause is a *resolution proof* on the original clauses. The 1-UIP resolution procedure is exactly backward chaining through the implication graph via resolution steps.

Worked CDCL Example

Consider formula F over variables x_1, x_2, x_3, x_4 with clauses:

$$c_1 = (x_1 \vee x_2), \quad c_2 = (\neg x_1 \vee x_3), \quad c_3 = (\neg x_2 \vee x_3), \quad c_4 = (\neg x_3 \vee x_4), \quad c_5 = (\neg x_3 \vee \neg x_4)$$

Level	Action	Propagation	Status
1	Decide $x_1 = 0$	$c_1 \Rightarrow x_2 = 1$; $c_3 \Rightarrow x_3 = 1$; $c_4 \Rightarrow x_4 = 1$; $c_5 = (0 \vee 0) -$ conflict	Conflict!
	1-UIP: resolve c_5 on x_4 with c_4; learned clause $(\neg x_3)$		Backjump to level 0
0	Unit prop: $x_3 = 0$	$c_2 \Rightarrow x_1 = 0$; $c_1 \Rightarrow x_2 = 1$; $c_3 = (0 \vee 0) -$ conflict	Conflict!
	Level 0 conflict $\Rightarrow$ **UNSAT**		

The formula is unsatisfiable. CDCL discovered this in two conflicts: the first at level 1 (learning $\neg x_3$), the second at level 0 (no further backtracking possible).

Non-Chronological Backtracking

Chronological (DPLL):

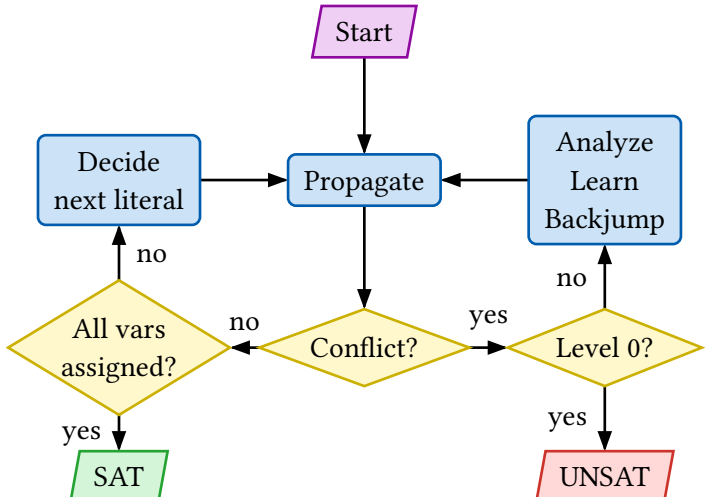
- Conflict at level k
- Go back to level $k - 1$
- Try the other branch
- May waste work if level $k - 1$ was irrelevant

Non-chronological (CDCL):

- Conflict at level k
- Analyze: learned clause has highest levels k and j ($j < k$)
- **Backjump** directly to level j
- Skip all levels between j and k

Backjumping skips irrelevant search space. A learned clause with highest levels k and j ($j < k$) means: once we have tried the level-k decision, we can jump straight to level j without exploring anything in between. Combined with clause learning, the solver never makes the same conflict-causing assignment twice.

CDCL Flowchart



CDCL Heuristics

Beyond the core algorithm, several heuristics make CDCL practical:

VSIDS (Variable State Independent Decaying Sum):

- Track an *activity score* per variable; bump the score of variables in each conflict clause.
- Periodically multiply all scores by a decay factor.
- Always pick the highest-activity unassigned variable next.
- Focuses search on variables that appear in recent conflicts — the structurally "hard" part.

Restarts:

- Periodically restart from scratch, keeping all learned clauses.
- Schedule via Luby sequence or geometric intervals.
- Escapes unproductive regions of the search space.

Phase saving:

- When a variable is decided, use its most recently assigned polarity.
- Quickly rediscovers satisfying sub-assignments after a restart.

SAT Solver Architecture

A modern CDCL solver (*e.g.*, MiniSat, 2k lines of C++) consists of:

Component	Purpose
Clause database	Stores original + learned clauses; periodically garbage-collects
Two-watched-literal scheme	Efficient unit propagation — only visit a clause when a watched literal becomes false
Decision heuristic (VSIDS)	Pick the next branching variable; bump activity on conflict
Conflict analysis (1-UIP)	Derive learned clause from implication graph
Restart policy	Luby or geometric; prevents getting stuck in unproductive subtrees
Phase saving	Remember last polarity of each variable for faster re-exploration

Note: The two-watched-literal scheme is the key to scalability — it makes unit propagation amortized $O(1)$ per propagation step, instead of scanning every clause.

Modern SAT Solvers and Competitions

Key solvers:

- **MiniSat** (Eén & Sörensson, 2003) — clean reference implementation, widely embedded in tools.
- **CaDiCaL** (Biere) — state-of-the-art, incremental API, DRAT proof logging.
- **Kissat** (Biere, 2020) — competition-optimized, inprocessing techniques.

SAT Competition (annual since 2002):

- **Industrial** track: verification and planning instances from industry.
- **Crafted** track: hard combinatorial benchmarks.
- **Random** track: random k-SAT near the phase transition.
- Open-source requirement drives solver improvement.

Modern solvers routinely handle *millions* of variables and *billions* of clauses. NP-complete in theory; polynomial in practice for structured industrial instances — the gap is entirely due to CDCL plus engineering.

Applications of SAT

Hardware Verification

- Bounded Model Checking (BMC): unroll circuit k times, check for bugs via SAT.
- Equivalence checking: are two circuits functionally identical?
- Used at Intel, AMD, ARM for chip design validation.
- *Intel Pentium FDIV bug (1994)*: cost $475M. Modern SAT-based verification would have caught it.

AI Planning

- Encode: can we reach goal state in k steps?
- SATPlan: competitive with dedicated planners.
- Mars rover path planning: 60-minute problems solved in seconds.

Software Analysis

- Symbolic execution backends (KLEE, SAGE).
- Concolic testing: concrete + symbolic execution.
- Configuration coverage: Linux kernel has 15k+ options $\Rightarrow 2^{15000}$ configs — SAT finds bugs in specific combinations.

Cryptanalysis & Mathematics

- *Boolean Pythagorean Triples theorem (2016)*: proved using SAT solver, generated 200 TB proof!
- Attack stream ciphers via algebraic SAT encoding.
- Factorization: $n = p \times q$ encoded as multiplication circuit + SAT.

Applications of SAT [2]

In 2016, researchers used a SAT solver to solve the Boolean Pythagorean Triples problem — a 35-year-old open problem in Ramsey theory. The solver ran for 2 days on 800 cores, exploring 10^{18} search states, and produced a 200 TB proof (largest math proof ever).

- *The problem:* Can we 2-color positive integers such that no Pythagorean triple $a^2 + b^2 = c^2$ is monochromatic?
- *Answer:* Yes up to 7824, impossible beyond.

CDCL: Key Takeaways

CDCL = DPLL + clause learning + non-chronological backtracking.

- Analyzes conflicts via *implication graphs*.
- Derives *learned clauses* that prune future search.
- *Backjumps* to the relevant decision level, skipping irrelevant levels.
- VSIDS + restarts + phase saving = practical efficiency.
- Basis of *every* competitive SAT solver since 2000.

Summary and Exercises

Summary

SAT Encoding:

- Variables capture binary choices
- ALO, AMO, EO constrain cardinality
- Implication $a \to b$ = one clause
- 4-step methodology: model $\Rightarrow$ constrain $\Rightarrow$ CNF $\Rightarrow$ solve

SAT Solving:

- DPLL: backtracking + unit propagation
- CDCL: + clause learning + backjumping
- Implication graphs trace propagation
- 1-UIP: derive learned clauses at conflicts
- VSIDS, restarts: practical performance

The pipeline: Problem $\to$ SAT encoding (ALO/AMO/EO) $\to$ DIMACS CNF $\to$ CDCL solver $\to$ model or UNSAT proof.
Next lecture: FOL theories and SMT — extending SAT with background knowledge about arithmetic, arrays, and uninterpreted functions.

Exercises: SAT Encoding

1. Encode the following problem as a SAT instance: *Schedule 4 lectures into 3 time slots such that no two lectures with a shared student occur in the same slot.* Define the variables, ALO/AMO/EO constraints, and conflict constraints.
2. Write a Python script to generate the DIMACS CNF encoding for vertex coloring of a graph with n vertices, m edges, and k colors. Test it on the Petersen graph ($n = 10$, $k = 3$).
3. Show that a DNF formula can be converted to an equivalent CNF in exponential size in the worst case, but the Tseitin encoding produces an *equisatisfiable* CNF of linear size. Why does equisatisfiability (rather than equivalence) suffice for SAT solving?

Exercises: DPLL

1. Run the DPLL algorithm (with unit propagation) on the formula: $(A \vee B \vee C) \wedge (\neg A \vee B) \wedge (\neg B \vee C) \wedge (\neg C \vee A) \wedge (\neg A \vee \neg B \vee \neg C)$ Draw the search tree showing all decisions, propagations, and backtracks.
2. Explain why the pure literal rule is *sound* (preserves satisfiability) but is rarely used in modern solvers.
3. $\star$ Consider the pigeonhole formula PHP_4^3 (4 pigeons, 3 holes). How many nodes does the DPLL search tree have in the worst case? What is the optimal variable ordering?

Exercises: CDCL

1. Given the following implication graph, identify the 1-UIP and derive the learned clause:
   Decisions: $x_1 = 1$ (level 1), $x_2 = 0$ (level 2).
   Propagations: $x_3 = 1$ (from $c_1 : \neg x_1 \vee x_3$), $x_4 = 1$ (from $c_2 : x_2 \vee x_4$), conflict on $c_3 : \neg x_3 \vee \neg x_4$.
2. Explain why a conflict at decision level 0 implies UNSAT.
3. Compare DPLL and CDCL on the formula from Exercise 1 above. Does CDCL learn any useful clauses?
4. $\star$ Show that every CDCL execution on an unsatisfiable formula implicitly constructs a *resolution proof*. Explain why this means CDCL can never be worse than tree-like resolution (up to polynomial overhead).

Bibliography

[1] M. Davis and H. Putnam, “A computing procedure for quantification theory,” *Journal of the ACM*, vol. 7, no. 3, pp. 201–215, 1960, doi: 10.1145/321033.321034.

[2] M. Davis, G. Logemann, and D. Loveland, “A machine program for theorem-proving,” *Communications of the ACM*, vol. 5, no. 7, pp. 394–397, 1962, doi: 10.1145/368273.368557.

[3] R. Nieuwenhuis, A. Oliveras, and C. Tinelli, “Solving SAT and SAT Modulo Theories: From an abstract Davis-Putnam-Logemann-Loveland procedure to DPLL(T),” *Journal of the ACM*, vol. 53, no. 6, pp. 937–977, 2006, doi: 10.1145/1217856.1217859.